

JOB MARKET ANALYSIS

PRDA-04 — Data Science Internship Project

Submitted By:
ALMUBEEN I

Course: Certified Data Analyst (CDA)

Organization: DataMites

Project: Job Market Analysis

Tools Used:

MySQL | Python | Pandas | Power BI

Dataset:

742 Job Postings | 42 Features

Year: 2026

DECLARATION

I hereby declare that the project titled “**Job Market Analysis**” has been completed as part of the DataMites Certified Data Analyst (CDA) internship project.

The analysis was performed using the provided job-market dataset and involved database querying, data preparation, exploratory analysis, visualization, and interpretation using MySQL, Python, Pandas, and Power BI.

The findings and recommendations presented in this report are based on the analysis performed on the available dataset.

Name: ALMUBEEN I

Course: Certified Data Analyst (CDA)

Project: PRDA-04 – Job Market Analysis

Year: 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **DataMites** for providing me with the opportunity to work on the Job Market Analysis project as part of the Certified Data Analyst internship program.

This project provided practical exposure to data extraction, SQL analysis, data cleaning, exploratory data analysis, visualization, and business interpretation.

I would also like to thank the trainers and project management team for providing the project requirements and technical guidance throughout the internship.

Finally, I would like to thank everyone who supported me during the completion of this project.

TABLE OF CONTENTS

1. Executive Summary
2. Introduction
3. Problem Statement
4. Project Objectives
5. Scope of the Project
6. Dataset Description
7. Tools and Technologies
8. Data Understanding
9. Data Cleaning and Preprocessing
10. SQL Analysis
11. Python Analysis
12. Power BI Dashboard
13. Key Insights
14. Business Recommendations
15. Limitations
16. Future Scope
17. Conclusion
18. Appendix

1. EXECUTIVE SUMMARY

The **Job Market Analysis** project focuses on understanding trends and patterns in Data Science-related employment opportunities using a dataset containing **742 job postings and 42 features**.

The dataset contains information related to job titles, salary estimates, company ratings, company names, locations, headquarters, company size, industry, sector, revenue, ownership type, required skills, job seniority, and educational qualifications. The assignment specifically

aims to identify in-demand skills, qualifications, job responsibilities, and market patterns that can support both job seekers and employers.

The project was completed using a combination of **MySQL, Python, Pandas, and Power BI**.

MySQL was used to retrieve and analyze the data through structured SQL queries. Python and Pandas were used for data extraction, preprocessing, exploratory analysis, statistical calculations, and generation of analytical tables and charts. Power BI was used to create an interactive dashboard for presenting the major findings.

The analysis examined job demand across locations, industries, companies, job titles, salaries, skills, education levels, seniority levels, company characteristics, and other available features.

The project provides a consolidated view of the Data Science job market and highlights areas that can help professionals understand where opportunities are concentrated, which skills are most frequently requested, and how compensation varies across different job-market characteristics.

2. INTRODUCTION

The growth of data-driven decision-making has increased the demand for professionals with skills in Data Science, Machine Learning, analytics, programming, cloud technologies, and business intelligence.

Organizations across different industries increasingly depend on data to understand customers, improve operations, forecast trends, automate processes, and support strategic decisions. As a result, understanding the characteristics of available Data Science jobs has become important for both employers and job seekers.

Job Market Analysis helps transform raw job-posting information into meaningful insights. Instead of examining individual job postings separately, analytical techniques can be used to identify broader patterns such as:

- Locations with high job demand
- Industries hiring Data Science professionals
- Companies with a large number of openings
- Frequently advertised job titles
- Salary differences across locations and roles
- Skills most frequently requested
- Relationship between education and salary
- Relationship between seniority and compensation

The PRDA-04 assignment provides a dataset containing **742 rows and 42 features** for this purpose.

This project uses SQL, Python, and Power BI to analyze these characteristics and present the results in a business-friendly format.

3. PROBLEM STATEMENT

The Data Science employment market contains a large variety of job titles, companies, industries, locations, salary ranges, skills, and qualification requirements.

However, raw job-posting data alone does not provide an immediate understanding of the overall market.

The key problem addressed by this project is:

How can job-posting data be analyzed to identify employment demand, salary patterns, skill requirements, qualification trends, and other factors that influence the Data Science job market?

The project aims to convert the available job-posting data into actionable information that can support:

- Job seekers planning their career development
- Employers understanding hiring patterns
- Training providers identifying important skills
- Analysts understanding employment-market trends

4. PROJECT OBJECTIVES

The major objectives of this project are:

1. Retrieve job-market data from the provided database.
2. Identify locations with the highest number of job postings.
3. Calculate average minimum and maximum salaries across locations.
4. Analyze average salaries across different locations.
5. Identify the top five industries with the highest number of Data Science-related job postings.
6. Identify companies with the highest number of job openings.
7. Identify the most frequently posted job titles.
8. Analyze salary levels for frequently posted job titles.
9. Analyze the skills required for different job titles.
10. Study the relationship between average salary and education.

11. Analyze the available features and derive additional insights.
12. Visualize the results using Power BI and provide business inputs and recommendations.

5. SCOPE OF THE PROJECT

The scope of the project covers the analysis of Data Science-related job postings contained within the provided dataset.

The analysis focuses on five major areas:

Job Demand

Understanding where jobs are concentrated and which companies and industries generate the highest number of opportunities.

Salary Analysis

Understanding salary ranges and comparing average compensation across locations, job titles, education levels, and seniority.

Skill Analysis

Identifying the technical skills most frequently requested by employers.

Qualification Analysis

Examining the relationship between educational qualifications and salary.

Business Intelligence

Using Power BI to transform analytical results into an interactive dashboard that allows users to explore the data.

6. DATASET DESCRIPTION

The provided dataset contains:

Attribute	Description
Number of Records	742
Number of Features	42
Domain	Data Science Job Market
Salary Currency	US Dollars
Salary Unit	\$K
Missing Value Indicator	-1

The assignment documentation states that the dataset contains 742 rows and 42 features. It includes job information such as job title, salary estimate, job description, company rating, company information, location, industry, sector, revenue, and other characteristics.

Major Feature Categories

Job Information

- Job Title
- Job Description
- Job Title Similarity
- Seniority

Salary Information

- Salary Estimate
- Lower Salary
- Upper Salary
- Average Salary

Company Information

- Company Name
- Headquarters
- Company Size
- Founded Year
- Ownership Type
- Revenue
- Rating

Market Information

- Location
- Industry
- Sector
- Competitors

Technical Skills

- Python
- Spark
- AWS
- Excel
- SQL
- SAS
- Keras

- PyTorch
- Scikit-learn
- TensorFlow
- Hadoop
- Tableau
- BI
- Flink
- MongoDB
- Google Analytics

Education

- Degree

The assignment specifies that skill fields use 1 when the skill is required and 0 when it is not required. It also states that -1 indicates unavailable information.

7. TOOLS AND TECHNOLOGIES

7.1 MySQL

MySQL was used for:

- Connecting to the project database
- Retrieving the Market table
- Understanding the database schema
- Performing aggregation
- Grouping job postings
- Calculating salary statistics
- Identifying top companies, industries, locations, and job titles

7.2 Python

Python was used for:

- Database connectivity
- Data extraction
- Data cleaning
- Exploratory Data Analysis
- Statistical analysis
- Generating analytical datasets
- Creating charts

7.3 Pandas

Pandas was used for:

- Loading database records
- DataFrame operations
- Missing-value analysis
- Grouping and aggregation
- Data transformation
- Exporting analytical results

7.4 Power BI

Power BI was used to create an interactive dashboard containing:

- KPI cards
- Job-location analysis
- Industry analysis
- Company analysis
- Job-title analysis
- Salary analysis
- Skill-demand analysis
- Education analysis
- Seniority analysis
- Interactive slicers

8. DATA UNDERSTANDING

Before performing detailed analysis, the structure and characteristics of the dataset were examined.

The database table used for the project is **Market**.

The table contains **42 columns**, covering job information, company characteristics, salary information, skills, education, and seniority.

The initial data inspection helped identify:

- Numeric columns
- Categorical columns
- Binary skill indicators
- Salary-related fields
- Missing/unavailable values
- Job and company attributes

This step was important because the assignment explicitly states that -1 values represent unavailable information and should be handled appropriately.

9. DATA CLEANING AND PREPROCESSING

Data cleaning was performed before the analytical stage.

9.1 Handling Unavailable Values

The dataset uses -1 to represent unavailable information.

Therefore, -1 was treated as an unavailable value rather than as a meaningful numerical value.

For salary calculations, unavailable salary values were excluded from average calculations.

9.2 Salary Preparation

The dataset contains:

- Lower Salary
- Upper Salary
- Average Salary

The assignment specifies that **Average Salary (K)** is expressed in thousands of US dollars.

Therefore, salary calculations were performed using the appropriate salary fields while excluding unavailable values.

9.3 Categorical Data

Fields such as:

- Job Location
- Industry
- Sector
- Degree
- Seniority
- Company Size
- Ownership Type

were analyzed as categorical variables.

9.4 Skill Fields

The technical-skill fields were represented using binary values:

- 1 → Skill required
- 0 → Skill not required

This allowed skill demand to be calculated by summing the number of job postings requiring each skill.

10. SQL ANALYSIS

SQL was used to answer the core questions defined in the PRDA-04 assignment.

10.1 Total Number of Job Postings

Result: 742 job postings

This confirms the total number of records available for analysis.

Interpretation

The dataset provides 742 individual job-posting records for analyzing Data Science employment trends.

10.2 Total Number of Features

Result: 42 features

The 42 variables provide multiple dimensions for analyzing the job market.

These include salary, company characteristics, job characteristics, skills, education, and seniority.

10.3 Job Postings by Location

The number of job postings was grouped by Job_Location.

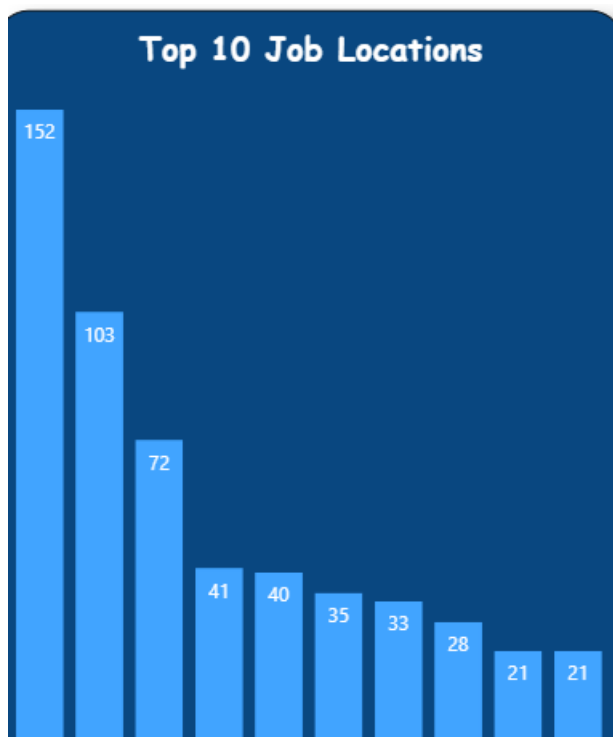
Power BI Visualization:

Top 10 Job Locations

Insert screenshot here.

Result

Top location:



Top location: California (CA)

Number of postings: 152

California has the highest number of Data Science job postings in the dataset, with 152 postings, indicating a strong concentration of opportunities in this location.

Interpretation

The distribution shows that Data Science job opportunities are concentrated in specific locations rather than being evenly distributed across all regions.

Insight

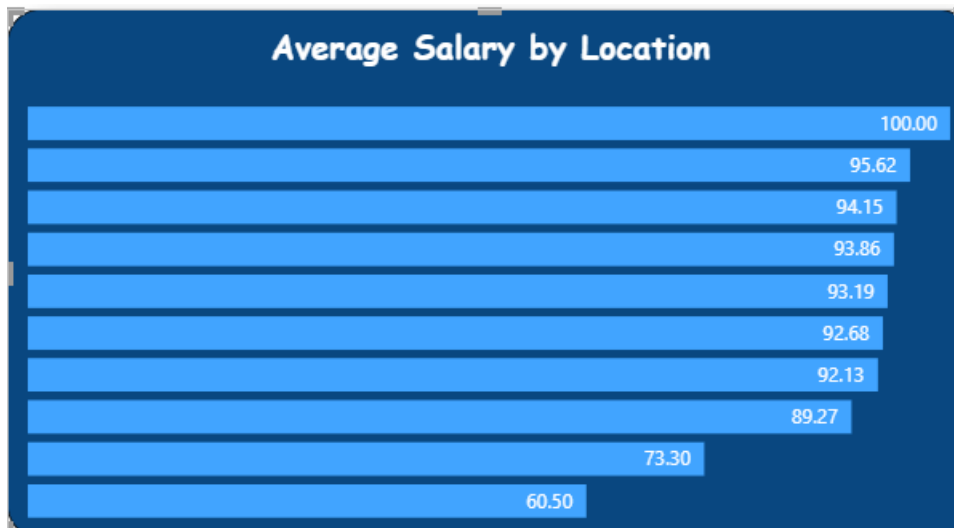
Locations with a higher concentration of job postings may represent stronger employment markets for Data Science professionals.

10.4 Average Salary by Location

The lower and upper salary values were analyzed for each location.

Visualization:

Average Salary by Location



Interpretation

Salary levels vary between locations, indicating differences in compensation across regional job markets.

Insight

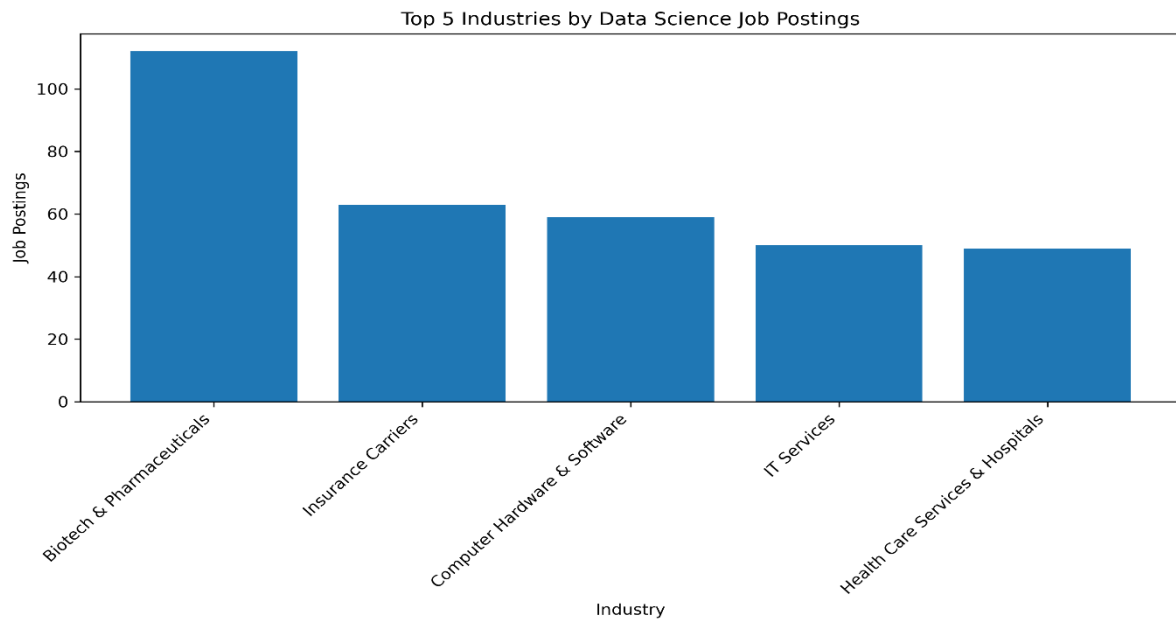
Job seekers can use location-based salary information when evaluating relocation opportunities or comparing offers.

10.5 Top Five Industries

The top five industries were identified based on the number of job postings.

Visualization:

Top 5 Industries



Interpretation

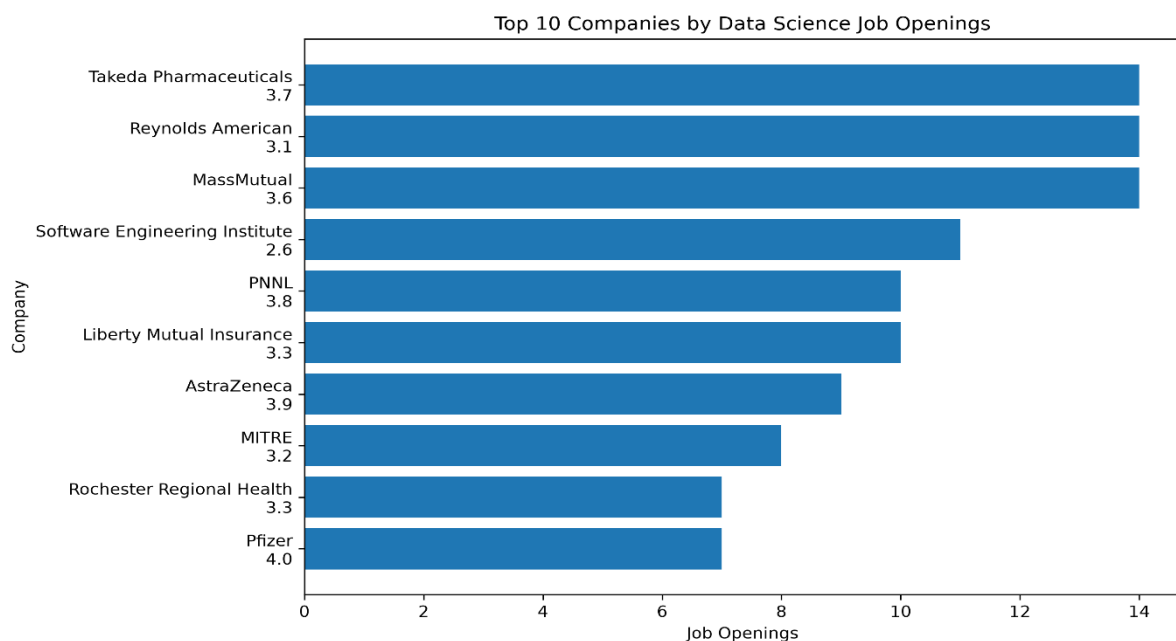
The concentration of job postings in these industries indicates where Data Science capabilities are most frequently being recruited.

10.6 Companies with Maximum Job Openings

Companies were grouped according to the number of job postings.

Visualization:

Top Companies by Job Openings



Interpretation

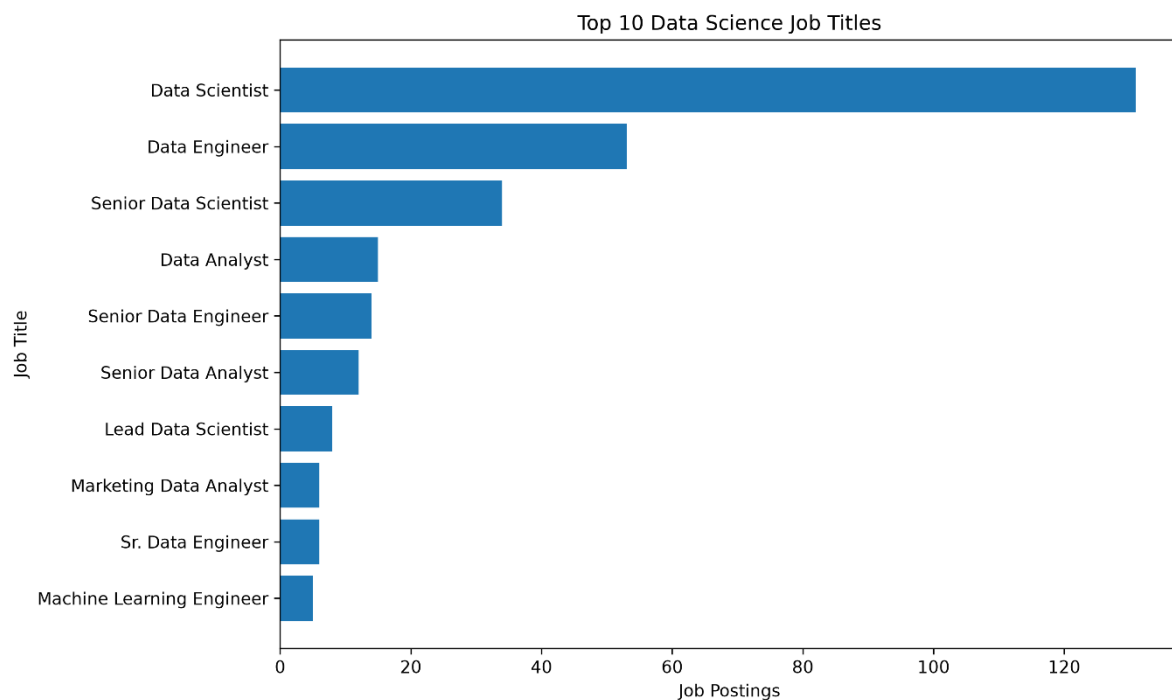
Companies with multiple job postings may have larger or more active Data Science hiring requirements.

10.7 Most Common Job Titles

Job titles were grouped and ranked by posting frequency.

Visualization:

Top Job Titles



Interpretation

Frequently appearing job titles indicate the roles most commonly represented in the dataset.

10.8 Salary by Job Title

Salary statistics were calculated for frequently posted job titles.

The analysis included:

- Average minimum salary



- Average maximum salary



- Average salary



Interpretation

Salary differences between job titles can help professionals understand how compensation varies across different Data Science roles.

11. SKILL ANALYSIS

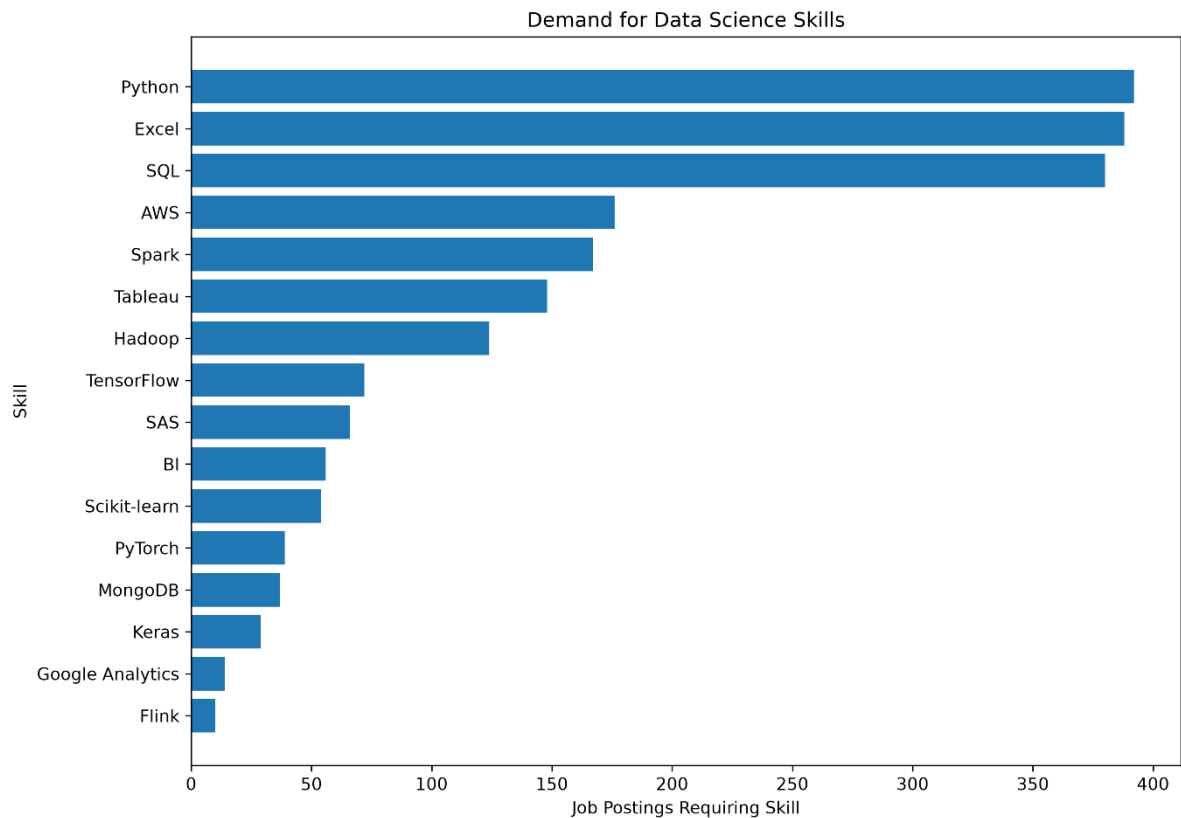
Technical skill requirements were analyzed using the binary skill columns.

The most frequently requested skills in the dataset were:

1. **Python — 392 postings**
2. **Excel — 388 postings**
3. **SQL — 380 postings**
4. **AWS — 176 postings**
5. **Spark — 167 postings**

Visualization:

Skill Demand



Key Insight

Python, Excel, and SQL are the most frequently requested skills in the analyzed job postings, with Python appearing in 392 postings, Excel in 388 postings, and SQL in 380 postings.

Recommendation for Job Seekers

Professionals targeting Data Science roles should prioritize developing strong skills in Python, Excel, SQL, AWS, and Spark.

- Python
- SQL
- Cloud platforms such as AWS
- Big-data technologies such as Spark

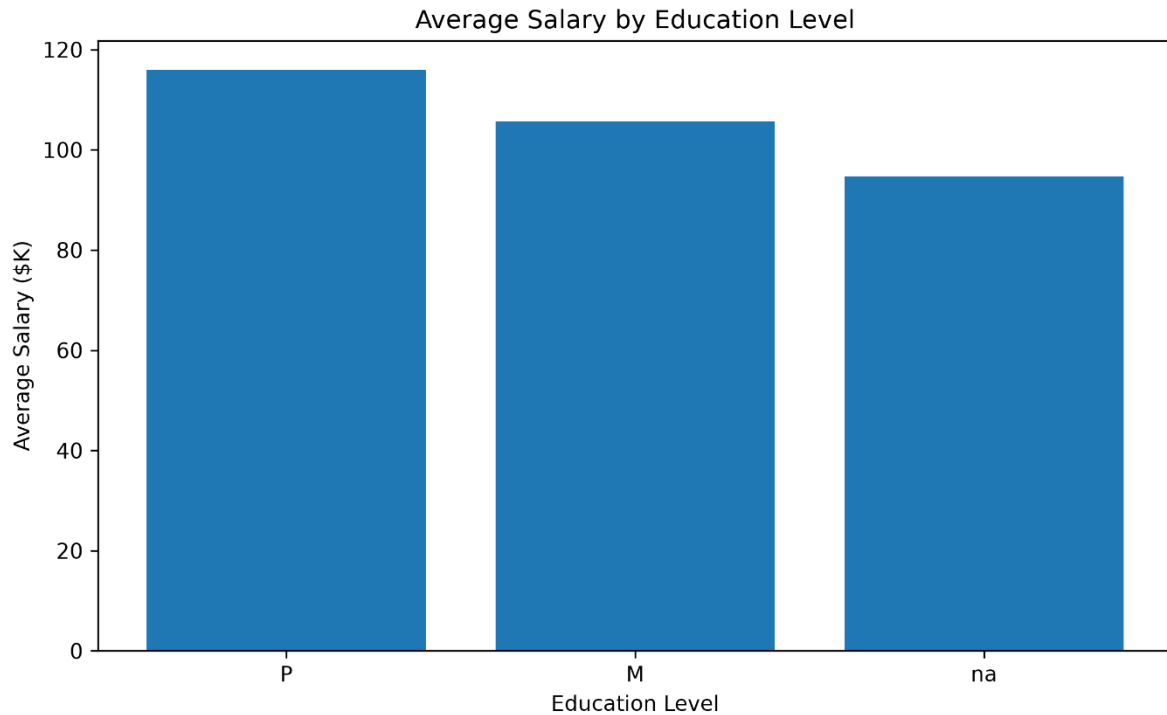
The assignment specifically identifies skill-demand analysis as one of the core objectives of the project.

12. EDUCATION AND SALARY ANALYSIS

The relationship between education and average salary was analyzed using the Degree field.

Visualization:

Average Salary by Education



The available categories include:

- Master's-related qualification (M)
- Ph.D.-related qualification (P)
- No specified degree information (na)

The assignment defines Degree as whether the job description mentions experience credit for a Master's or Ph.D. degree.

Interpretation

The salary distribution across education categories indicates whether advanced educational qualifications are associated with differences in compensation within the dataset.

Insight

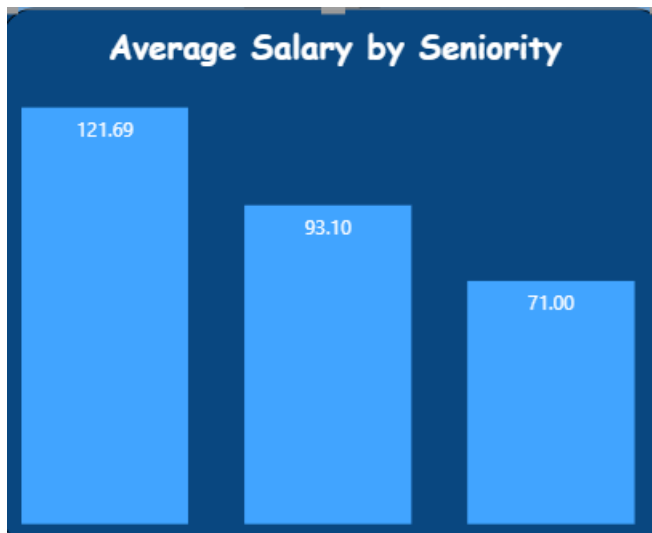
Education may influence salary expectations, but it should be considered alongside job title, seniority, skills, location, and industry.

13. SENIORITY AND SALARY ANALYSIS

Job seniority was analyzed using the seniority_by_title field.

Visualization:

Average Salary by Seniority



Interpretation

The analysis helps identify how compensation varies across different seniority categories.

Insight

Differences in salary between seniority levels can help job seekers understand the potential compensation progression associated with career advancement.

14. ADDITIONAL FEATURE ANALYSIS

To satisfy the requirement to analyze all available features and derive multiple insights, additional dimensions were examined.

These included:

Company Size

Analyzed job-posting volume and average salary across company-size categories.

Industry

Compared job demand and average compensation across industries.

Sector

Compared job postings and salary levels across sectors.

Ownership Type

Analyzed job postings across private, public, government, and other ownership categories where available.

Company Rating

Examined the relationship between company rating and job-posting volume and salary.

Revenue

Compared job postings and salary across company-revenue ranges.

Employer-Provided Salary

Compared job postings where salary information was employer-provided versus other postings.

Hourly vs Non-Hourly Positions

Compared job postings and salary between hourly and non-hourly positions.

Company Age

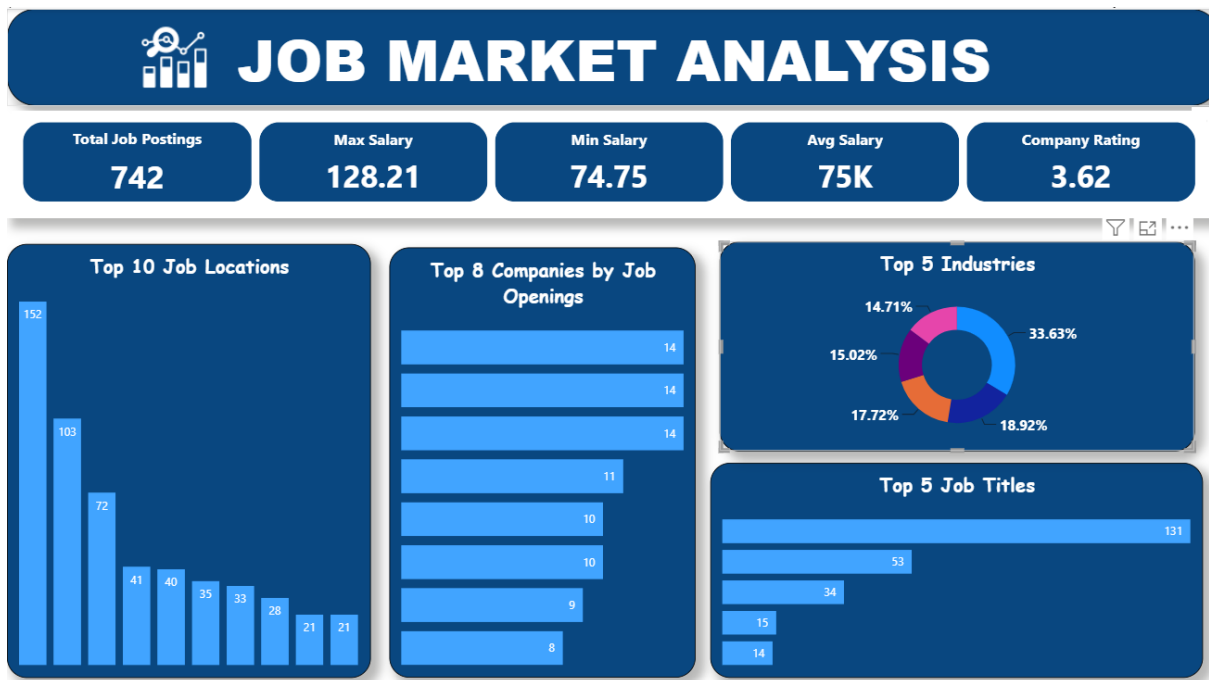
Grouped companies by age and analyzed job-posting volume and average salary.

These analyses provide a broader understanding of the factors associated with Data Science employment opportunities.

15. POWER BI DASHBOARD

The Power BI dashboard was developed using two pages.

Page 1 — Job Market Overview



The first dashboard page provides a high-level overview of the job market.

KPI Cards

The dashboard includes:

- **Total Job Postings — 742**
- **Max Salary — 128.21K**
- **Min Salary — 74.75K**
- **Avg Salary — 75K**
- **Company Rating — 3.62**

Visualizations

- Top 10 Job Locations
- Top Companies by Job Openings
- Top 5 Job Titles
- Top 5 Industries

Interactive Filters

- Job Location
- Industry

These slicers allow users to interactively filter the dashboard.

16. PAGE 2 — SALARY & SKILLS ANALYSIS



The second dashboard page focuses on salary, skills, education, and seniority.

Visualizations

1. Average Salary by Location
2. Skill Demand
3. Average Salary by Education
4. Average Salary by Seniority

Interactive Filter

- Education Level

This page allows users to examine compensation and skill patterns in greater detail.

17. KEY INSIGHTS

Based on the analysis performed, several important observations can be derived.

Insight 1 — Job Demand is Concentrated

The job postings are not evenly distributed across locations. A relatively small number of locations account for a significant share of available opportunities.

Business**implication:**

Job seekers targeting high-demand locations may have access to a larger pool of opportunities.

Insight 2 — Data Science Salaries Show Regional Variation

Average salary differs across locations.

Business implication:

Location should be considered when evaluating job offers because compensation expectations can vary across regional markets.

Insight 3 — Python is Highly Demanded

Python is required in **392 job postings**, making it one of the most frequently requested technical skills in the dataset.

Recommendation:

Data Science professionals should maintain strong Python programming and analytical capabilities.

Insight 4 — Big Data and Cloud Skills are Important

Spark appears in **388 postings**, while AWS appears in **380 postings**.

Recommendation:

Combining Data Science fundamentals with big-data and cloud technologies can improve alignment with employer requirements.

Insight 5 — Salary Varies Across Seniority

The seniority analysis shows differences in average compensation across different position levels.

Recommendation:

Professionals should consider career progression and experience development when planning their long-term Data Science careers.

Insight 6 — Education is One of Several Salary Factors

The education analysis shows differences in average salary between education categories.

However, education should not be considered independently because salary can also depend on:

- Job title
- Seniority
- Skills
- Location
- Industry
- Company characteristics

18. BUSINESS RECOMMENDATIONS

18.1 Recommendations for Job Seekers

Build Strong Python Skills

Python appears among the most frequently requested skills in the dataset.

Develop Cloud Skills

AWS is highly represented in job postings, indicating the importance of cloud capabilities.

Learn Big-Data Technologies

Spark is also highly demanded, making big-data processing an important complementary skill.

Strengthen SQL

SQL remains important for working with structured data and databases.

Consider Location

Professionals should compare salary levels and job demand across locations before making relocation or job-selection decisions.

Develop Senior-Level Capabilities

Professionals should build experience and advanced technical capabilities to progress toward higher-seniority roles.

18.2 Recommendations for Employers

Focus on High-Demand Skill Combinations

Organizations should clearly define required technical skills in job descriptions based on actual role requirements.

Improve Job Description Quality

Clear job descriptions can help attract candidates with the appropriate skills and experience.

Monitor Market Salaries

Organizations should regularly benchmark compensation against relevant locations and roles to remain competitive.

Develop Internal Talent

Organizations can invest in employee training for high-demand skills such as Python, cloud technologies, SQL, and big-data technologies.

18.3 Recommendations for Training Providers

Training programs should prioritize skills that appear frequently in job advertisements.

A practical curriculum could include:

Python → SQL → Statistics → Machine Learning → AWS → Spark → Data Visualization

This sequence provides a combination of foundational, analytical, engineering, cloud, and visualization skills.

19. LIMITATIONS

The project has several limitations.

Dataset Size

The analysis is based on 742 job postings and may not represent the entire Data Science employment market.

Missing Information

The dataset contains -1 values representing unavailable information. The assignment explicitly identifies -1 as unavailable data.

Historical Nature

Job-market conditions can change over time. Therefore, the findings should be interpreted as observations from the provided dataset rather than permanent market conditions.

Salary Reporting

Salary estimates may vary by source and may not represent the final compensation received by an employee.

Job Posting Bias

A job posting represents an advertised opportunity and does not necessarily represent the complete hiring activity of an organization.

20. FUTURE SCOPE

Future versions of this project could include:

- Real-time job-market data
- Time-series analysis of job demand
- Job-posting trend prediction
- Salary prediction using Machine Learning
- Skill-demand forecasting
- Natural Language Processing of job descriptions
- Automated job recommendation systems
- Candidate-job matching
- Geographic visualization using maps
- Interactive Power BI drill-through reports

Machine Learning could also be used to predict salary based on:

- Location
- Experience
- Education
- Skills
- Industry
- Company characteristics
- Seniority

21. CONCLUSION

The **Job Market Analysis** project demonstrates how structured data-analysis techniques can be used to understand the Data Science employment market.

The project analyzed **742 job postings across 42 features**, covering job characteristics, salaries, companies, locations, industries, skills, education, and seniority.

MySQL provided the foundation for structured data analysis, while Python and Pandas supported data preparation, exploratory analysis, and analytical output generation. Power BI transformed the analytical results into an interactive dashboard.

The analysis identified important patterns in job demand, salary, technical skills, education, and seniority. In particular, Python, Spark, and AWS appeared among the most frequently requested technical skills in the analyzed postings.

Overall, the project demonstrates the value of combining **SQL, Python, and Business Intelligence tools** to transform raw job-market data into actionable insights.

The findings can help job seekers understand market expectations, employers improve hiring strategies, and training providers design programs around skills that are frequently requested in Data Science job postings.

22. APPENDIX

Appendix A — SQL Queries

- Total job count
- Number of features
- Location analysis
- Salary analysis
- Industry analysis
- Company analysis
- Job-title analysis
- Skill analysis
- Education analysis
- Seniority analysis
- Additional feature analysis
- Data-quality analysis

Appendix B — Python Analysis

Include:

- Python script
- Data extraction process
- Cleaning process
- Analysis-table generation
- Chart-generation process

Appendix C — Power BI

Include:

- Page 1 screenshot
- Page 2 screenshot

Appendix D — Generated Analysis Files

Mention the analytical outputs generated during the project:

- job_market_data.csv
- missing_values.csv
- location_analysis.csv
- salary_by_location.csv
- top_5_industries.csv
- top_companies.csv
- top_job_titles.csv
- skill_demand.csv
- education_vs_salary.csv
- seniority_vs_salary.csv
- industry_vs_salary.csv
- sector_vs_salary.csv
- company_size_vs_salary.csv
- ownership_vs_salary.csv
- rating_vs_salary.csv
- revenue_vs_salary.csv
- employer_provided_salary.csv
- hourly_vs_salary.csv
- company_age_analysis.csv